

Adam Bouafia

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Summary

Cloud and DevOps engineer with an MSc in Computer Science (double degree, VU Amsterdam and Università dell'Aquila). Hands-on experience with Azure, Kubernetes, and CI/CD pipelines, including production Azure infrastructure built entirely with Infrastructure as Code. Working through the Linux Foundation Advanced Cloud Engineer program and preparing for the CKA. Focused on automation, observability and green computing.

Experience

Master's Thesis Intern, Vrije Universiteit Amsterdam – Amsterdam, Netherlands Mar 2025 – Jan 2026

- Thesis: LogPress
- Semantic Log Compression System
- Processed 2.4M+ log records at a 22.33x compression ratio with 100% lossless reconstruction, cutting the storage and transfer footprint of log retention
- Built a Lazy Reconstruction query layer answering in 21-111ms without full decompression; containerised with Docker and released through GitHub Actions

DevOps Engineer, Vivieb – Pescara, Italy Nov 2023 – Feb 2024

- Optimised Kubernetes cluster energy efficiency using Kubegreen, measured through Grafana and InfluxDB dashboards
- Implemented automated deployment pipelines for the platform team

Software Developer, Euromed Organization – Paris, France July 2022 – Dec 2022

- Developed the Dhayen mobile security app for a national awareness campaign, with a CI/CD pipeline for releases

Education

Vrije Universiteit Amsterdam / Università dell'Aquila, MSc in Computer Science (Double Degree)

- Specialisation: Software Engineering and Green IT

Higher Institute of Computer Science of Mahdia, BSc in Software Engineering & Information System

Professional Development

The Linux Foundation - Advanced Cloud Engineer IT Professional Program Mar 2026 – present

- Eight-course cloud native track (LFS253, LFS258, LFS241, LFS242, LFS243, LFS244); built and operated a three-node kubeadm cluster on libvirt/KVM including upgrades, etcd backup and restore and control plane recovery; preparing for the CKA exam

Microsoft Azure Certification Path In progress

- Preparing for AZ-104 (Azure Administrator Associate), then AZ-400 toward the DevOps Engineer Expert credential

Projects

FirstCall - AI First Responder for Kubernetes Incidents Sep 2026

- Built for the AI Accel hackathon; ships as a Helm chart, collects first-15-minutes incident evidence and redacts secrets before anything leaves the cluster
- Read-only by default: fixes are allow-listed, server dry-run and applied only after human approval

repo-sentinel - Autonomous Repository Auditor Sep 2026

- Agent auditing repositories for failing CI, outdated dependencies and upstream drift, opening guarded fix PRs with no shell and no local file access

Open Source: Kubernetes-Certified-Administrator (walidshaari, 4.4k stars) Sep 2026

- Updated the CKA study guide to the 2026 curriculum (Kubernetes v1.35); merged upstream as PR #63 and added as a repository contributor

Azure Landing Zone - Production Infrastructure as Code Feb 2026

- Hub-spoke Azure topology in 100% Bicep (Firewall, Bastion, NSGs, Key Vault with RBAC), deployed via a multi-stage Azure DevOps pipeline with what-if preview and approval gates
- Cost automation reducing development infrastructure spend by 70%

Greenmining - Green IT Research Library Jan 2026

- Python library on PyPI implementing 124 Green Software Foundation sustainability patterns, with multi-backend energy measurement (RAPL, CodeCarbon) and carbon footprint reporting

CO2 Consumption Monitor - GNOME Shell Extension Sep 2025

- Published on the GNOME Extensions store: real-time CO2 per application, live grid intensity via ElectricityMaps, CSV export and an automated release pipeline

Think Before You Save - Green Waves Hackathon Winner Jan 2025

- Winning entry at the Green Waves hackathon run with ilionx and TU Delft: a browser extension showing the CO2 cost of cloud storage at the moment of saving

Skills

Green IT: Energy measurement (RAPL, CodeCarbon), ElectricityMaps, Kubegreen, Green Software Foundation patterns

Cloud & IaC: Azure, Bicep, Terraform, Helm

Containers & Orchestration: Kubernetes (kubeadm, RBAC, etcd operations), Docker, Helm, Knative

CI/CD: Azure DevOps, GitHub Actions, GitLab CI, Jenkins, ArgoCD

Observability: Prometheus, Grafana, InfluxDB, ELK Stack, Fluentd, Fluent Bit

Languages & Systems: Python, Bash, Java, JavaScript, SQL; Linux (Fedora, Ubuntu, RHEL); Git

Languages

English: Fluent

Arabic: Native

French: Proficient

Italian: Proficient

Dutch: A2, moving to B1 in October 2026